Vasu Ramesh Gadde

Applied AI Engineer | Generative AI | RAG | Document Intelligence & AI Automation

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PROFILE

Applied AI Engineer with 2+ years of combined experience developing Generative AI applications, document intelligence pipelines, and workflow automation solutions for financial and HR operations. Experienced in building hybrid OCR and LLM-based document processing systems, Retrieval-Augmented Generation (RAG) chatbots, conversational AI applications, and backend services using Python, FastAPI, and MongoDB. Hands-on experience integrating local and cloud-based LLMs, including Qwen2.5 and GPT-4o, for structured data extraction, document validation, information retrieval, and business process automation. Familiar with AWS and Docker-based deployment workflows.

PROFESSIONAL EXPERIENCE

JHS Associates & LLP, *Data Scientist & AI Engineer* 07/2025 – Present | Mumbai, India

- Developed a hybrid OCR and LLM-based invoice classification pipeline using Surya OCR and local Qwen2.5, with OpenAI API fallback to support document processing and improve output stability.
- Resolved production dependency conflicts within the document processing pipeline to stabilize system behavior.
- Automated payroll review and spreadsheet reconciliation workflows using Python, Pandas, and an OPE approval PDF export system.
- Developed a Compliance Signals Intelligence tool using FastAPI, MongoDB, and GPT-4o to identify and surface risk signals from business documents.
- Worked on an LLM-based document validation system with pluggable model providers, including Hugging Face and Groq.
- Restructured the JHS Timesnap HR platform into a modular FastAPI architecture to improve backend organization and maintainability.
- Authored technical build documentation for a Vendor Payment Management System covering 9 modules and 14 MongoDB schemas.
- Built browser automation using Playwright to download compliance reports from web-based sources.
- Developed a financial statement extraction pipeline using pdfplumber to support structured extraction from PDF documents.

Supermoney Advisors Private Limited, *Machine Learning Intern* 08/2024 – 07/2025 | Mumbai, India

- Developed business-specific conversational AI systems using the OpenAI API and FastAPI for internal analytics and question-answering use cases.
- Designed intent-aware prompt workflows and retrieval-based response generation to support relevant answers to user queries.
- Built data ingestion workflows using MongoDB and REST APIs to provide application data for chatbot responses.
- Integrated AI services with backend systems to support conversational interfaces and business-specific information retrieval.
- Deployed AI services on AWS using Docker to package and run application components.

EDUCATION

BSc in Information Technology, *University of Mumbai* 09/2021 – 05/2024 | Thane, India

SKILLS

Programming & Data Processing
Python, SQL, Pandas, NumPy

RAG & Information Retrieval
Retrieval-Augmented Generation (RAG), Embeddings, Vector Search, FAISS, Pinecone, Document Chunking, Semantic Search

Cloud & DevOps
AWS, Docker, GitHub Actions

Document Intelligence & Automation
OCR, Surya OCR, PDF Extraction, pdfplumber, Invoice Classification, Structured Data Extraction, Playwright

Generative AI & LLMs
Generative AI, Large Language Models (LLMs), OpenAI API, GPT-4o, Qwen2.5, Prompt Engineering, Agentic AI Workflows

Backend Development & Databases
FastAPI, REST APIs, MongoDB, SQL

AI Application Development
Conversational AI, Intent Classification, LLM-Based Validation, Workflow Automation

COURSES

Data Science Certification, *HCL GUVI* 03/2025 – 05/2026

- Completed a Data Science programme covering Python programming, SQL, statistics, exploratory data analysis, data visualization, and machine learning.
- Gained practical experience in data preprocessing, data analysis, visualization, and data-driven problem-solving through hands-on projects.

PROJECTS

AI Invoice & PDF Extraction (Document Intelligence), *Technologies: Python, FastAPI, Surya OCR, Qwen2.5, OpenAI API*

- Built a hybrid OCR and LLM pipeline to extract structured information from invoice PDFs.
- Integrated Surya OCR with local Qwen2.5 and cloud-based OpenAI API fallback for document processing.
- Implemented fallback logic between local and cloud LLMs to balance processing cost, latency, and extraction quality.
- Designed the workflow to support structured data extraction and invoice classification from unstructured documents.

RAG-Based Chatbot Systems, *Technologies: Python, FastAPI, OpenAI API, FAISS, Pinecone, MongoDB*

- Developed Retrieval-Augmented Generation chatbots using document ingestion, text chunking, embeddings, and vector search.
- Integrated retrieval pipelines with LLMs to generate responses grounded in relevant document or knowledge-base content.
- Worked on a crypto research agent combining RAG with LSTM-based forecasting.
- Developed a conversational agent with query-routing capabilities to support different types of user requests.

AI-Powered Recruitment Automation Platform, *Technologies: Python, FastAPI, OpenAI API, Pinecone, MongoDB*

- Built a Generative AI-based recruitment solution for resume parsing and structured candidate profile extraction.
- Implemented semantic candidate-job matching to evaluate the relevance of candidate profiles against job requirements.
- Designed backend APIs for resume storage, retrieval, and candidate suitability analysis.
- Used vector search and LLM-based processing to support candidate discovery and recruitment workflow automation.